

■ Hardware Components List

All components required to build the Automatic Exhaust Fan

Complete Hardware Components

The following table lists all hardware components used in this project along with their specifications, quantity, and purpose.

#	Component	Specification / Model	Qty	Purpose
1	ESP8266 NodeMCU V3	Wi-Fi MCU, 3.3V logic, 80MHz	1	Main microcontroller
2	MQ-2 Gas Sensor	Detects smoke, LPG, CH4, CO	1	Smoke detection
3	1-Ch Relay Module	5V coil, 10A/250VAC contacts	1	Fan ON/OFF switch
4	DC Motor (Fan)	4.5V–9V, 9000 RPM, 130-type	1	Exhaust fan
5	Active Buzzer	5V, piezo buzzer	1	Smoke alarm
6	Red LED	5mm, 2V forward voltage	1	Hazard indicator
7	Green LED	5mm, 2V forward voltage	1	Safe indicator
8	Orange LED	5mm, 2V forward voltage	1	Warning indicator
9	Resistors	220Ω, 1/4W	3	Current limiting for LEDs
10	5V Power Supply	DC adapter / USB 5V 1A	1	Power for NodeMCU
11	Jumper Wires	M-M and M-F, 20cm	20+	Connections
12	Breadboard	830-tie point	1	Prototyping

Key Component Details

ESP8266 NodeMCU V3

- Operating Voltage: 3.3V (5V tolerant via USB)
- CPU: 80MHz Tensilica L106
- Flash: 4MB
- GPIO Pins: 11 digital, 1 analog (A0)
- Built-in Wi-Fi: 802.11 b/g/n
- Programming: Arduino IDE / MicroPython

MQ-2 Gas Sensor

- Supply Voltage: 5V DC
- Detects: Smoke, LPG, CO, Methane, Butane, Propane

- Analog output: 0–5V (proportional to gas concentration)
- Digital output: HIGH when gas exceeds potentiometer threshold
- Warm-up time: ~20 seconds
- Sensitivity adjustment: via onboard potentiometer

5V 1-Channel Relay Module

- Coil voltage: 5V DC
- Contact rating: 10A at 250VAC / 10A at 30VDC
- Trigger: LOW signal from microcontroller
- Optocoupler isolated input
- LED indicator for relay status

DC Motor (Exhaust Fan)

- Type: Standard 130-type DC motor
- Operating Voltage: 4.5V to 9V
- No-load current: 70mA max
- No-load speed: 9000 RPM
- Loaded current: 250mA approx
- Size: 27.5mm x 20mm x 15mm, Weight: 17g